

Heisenberg quantum channels are operator-norm closed

A full answer to the closure question in arXiv:2012.03496

Run `fa_banach_001`, lane 1 (GPT5.6)

11 August 2026

Status: candidate full result, likely valid. For separable complex Hilbert spaces $\mathcal{G}, \mathcal{H}$, the set $Q_H(\mathcal{G}, \mathcal{H})$ is closed in $\mathcal{B}(\mathcal{B}(\mathcal{G}), \mathcal{B}(\mathcal{H}))$ for the operator norm.

1 Source and exact question

Vom Ende's thesis [1] defines a Schrödinger quantum channel as a completely positive trace-preserving map

$$T : \mathcal{B}^1(\mathcal{H}) \longrightarrow \mathcal{B}^1(\mathcal{G})$$

and a Heisenberg quantum channel as an ultraweakly continuous completely positive unital map

$$S : \mathcal{B}(\mathcal{G}) \longrightarrow \mathcal{B}(\mathcal{H}).$$

Their respective sets are denoted $Q_S(\mathcal{H}, \mathcal{G})$ and $Q_H(\mathcal{G}, \mathcal{H})$.

Immediately after Proposition 2.3.13, the source asks whether $Q_H(\mathcal{G}, \mathcal{H})$ is closed in the *full* bounded-map space $\mathcal{B}(\mathcal{B}(\mathcal{G}), \mathcal{B}(\mathcal{H}))$ [1, p. 48]. The exact source passage is reproduced in Figure 1.

2. Preliminaries

It remains unknown whether $Q_H(\mathcal{G}, \mathcal{H})$ is closed as a subset of $\mathcal{B}(\mathcal{B}(\mathcal{G}), \mathcal{B}(\mathcal{H}))$. If true it cannot be a mere corollary of Prop. 2.3.13 (ii) as one can find general examples where the weak*-operator topology does not transfer weak*-continuity to the limit (Remark 2.1.28).

Remark 2.3.14. The example which shows that $Q_S(\mathcal{H}, \mathcal{G})$ is not w.o.t.-compact is, after slight modification, the same which shows that $\mathbb{D}(\mathcal{H})$ is not weakly compact in infinite dimensions.

Perhaps the most important property of quantum channels for the purpose of analyzing reachable sets is the fact that every channel is a contraction:

Proposition 2.3.15. *Let $T \in Q_S(\mathcal{H}, \mathcal{G})$ and $S \in Q_H(\mathcal{G}, \mathcal{H})$. Then $\|T\| = 1$ and $\|S\| = 1$.*

Proof. As each $S \in Q_H(\mathcal{G}, \mathcal{H})$ is linear, positive, and unital it has operator norm $\|S\| = 1$ as a

Figure 1: The operator-norm closure question, arXiv:2012.03496v2, printed page 48 (physical PDF page 62).

2 Proof intuition

The source already identifies the two channel pictures by an isometric bijection: take the Banach-space adjoint. It also proves that the Schrödinger channel set is operator-norm closed, hence complete. An isometry preserves Cauchy sequences and limits, so its image—the entire Heisenberg channel set—is complete in the ambient operator norm. Complete subsets of metric spaces are closed. The weak-star continuity concern raised after the question never enters, because the norm-isometric preadjoint correspondence constructs the limiting normal map directly.

3 Formal proof

Lemma 1. *Let X be a complete metric space, Y a metric space, and $J : X \rightarrow Y$ an isometry. Then $J(X)$ is closed in Y .*

Proof. If $J(x_n) \rightarrow y$ in Y , then $(J(x_n))$ is Cauchy. Isometry makes (x_n) Cauchy, hence $x_n \rightarrow x$ for some $x \in X$. Again by isometry, $J(x_n) \rightarrow J(x)$, and uniqueness of metric limits gives $y = J(x)$. $\square$

Theorem 1. *Let $\mathcal{G}, \mathcal{H}$ be separable complex Hilbert spaces. Then*

$$Q_H(\mathcal{G}, \mathcal{H}) \text{ is operator-norm closed in } \mathcal{B}(\mathcal{B}(\mathcal{G}), \mathcal{B}(\mathcal{H})).$$

Proof. The source’s Corollary 2.3.12 says that every $S \in Q_H(\mathcal{G}, \mathcal{H})$ has the form $S = T^*$ for a unique $T \in Q_S(\mathcal{H}, \mathcal{G})$. The paragraph preceding Proposition 2.3.13, together with its footnote 33, says that

$$J : Q_S(\mathcal{H}, \mathcal{G}) \longrightarrow Q_H(\mathcal{G}, \mathcal{H}), \quad J(T) = T^*,$$

is a surjective isometry for the operator norms [1, pp. 46–47].

Proposition 2.3.13(i) states that $Q_S(\mathcal{H}, \mathcal{G})$ is operator-norm closed in

$$\mathcal{B}(\mathcal{B}^1(\mathcal{H}), \mathcal{B}^1(\mathcal{G}))$$

[1, p. 47]. Consequently $Q_S(\mathcal{H}, \mathcal{G})$ is complete. Lemma 1, applied to the displayed isometry with codomain the full space $\mathcal{B}(\mathcal{B}(\mathcal{G}), \mathcal{B}(\mathcal{H}))$, shows that its image is closed there. Surjectivity identifies that image with $Q_H(\mathcal{G}, \mathcal{H})$. $\square$

The two decisive source inputs are shown together in Figure 2.

4 Independent sequence audit

For clarity, suppose $S_n \in Q_H(\mathcal{G}, \mathcal{H})$ and $\|S_n - S\| \rightarrow 0$ in the full mapping space. Write $S_n = T_n^*$ using Corollary 2.3.12. Then

$$\|T_n - T_m\| = \|T_n^* - T_m^*\| = \|S_n - S_m\|,$$

so T_n converges in operator norm to some T . Proposition 2.3.13(i) gives $T \in Q_S$. Moreover $T_n^* \rightarrow T^*$ in norm, while $T_n^* = S_n \rightarrow S$. Thus $S = T^* \in Q_H$. This repeats the completeness proof without hiding any topological interface.

In other words this corollary tells us that the Schrödinger and the Heisenberg picture are equivalent as $Q_S(\mathcal{H}, \mathcal{G}) \simeq Q_H(\mathcal{G}, \mathcal{H})$ by means of the isometric isomorphism³³ $*$. Thus the structure of the set of all physical transformations does not change when switching the picture. Further algebraic and topological properties of the set of channels read as follows:

Proposition 2.3.13. *The following statements hold.*

- (i) *The set $Q_S(\mathcal{H})$ is a convex subsemigroup of $\mathcal{B}(\mathcal{B}^1(\mathcal{H}))$ with unity element $\mathbb{1}_{\mathcal{B}^1(\mathcal{H})}$. Moreover, $Q_S(\mathcal{H}, \mathcal{G})$ is closed in $\mathcal{B}(\mathcal{B}^1(\mathcal{H}), \mathcal{B}^1(\mathcal{G}))$ with respect to the weak operator, strong operator, and operator norm topology. However, as soon as $\mathcal{G}$ is of infinite dimension, then $Q_S(\mathcal{H}, \mathcal{G})$ is not compact in either of these topologies.*
- (ii) *The set $Q_H(\mathcal{H})$ is a convex subsemigroup of $\mathcal{B}(\mathcal{B}(\mathcal{H}))$ with unity element $\mathbb{1}_{\mathcal{B}(\mathcal{H})}$. Moreover, $Q_H(\mathcal{G}, \mathcal{H})$ is closed in $*(\mathcal{B}(\mathcal{B}^1(\mathcal{G}), \mathcal{B}^1(\mathcal{H})))$ with respect to the ultraweak operator, weak operator, strong operator, and operator norm topology. However, as soon as $\mathcal{G}$ is of infinite dimension, then $Q_H(\mathcal{G}, \mathcal{H})$ is not compact in either of these topologies.*

Proof. (i): Convexity, semigroup property, and closedness are due to [88, Thm. 1]. The

Figure 2: The source states both the isometric Schrödinger–Heisenberg identification and operator-norm closedness of Q_S (printed page 47).

5 Verification, limitations, and novelty

The proof uses only three statements explicitly present on adjacent source pages: surjectivity of the adjoint correspondence, its norm-isometry, and norm-closedness of Q_S . No finite-dimensional approximation, compactness, or weak-star limit argument is used. Hence the source’s warning that weaker operator topologies may fail to preserve weak-star continuity does not affect the theorem.

The claim is restricted to operator-norm closure and to the source’s separable Hilbert-space setting. It does not assert compactness or closure for weaker topologies.

On 11 August 2026, exact searches of the run registry and the solution, attempt, and proof-gap indexes found no packet for arXiv:2012.03496. Bounded primary-arXiv searches used the exact closure sentence, “Heisenberg quantum channels operator norm closed,” and “normal unital completely positive maps norm closed.” They found the source and later quantum-control work, but no paper explicitly recording this corollary. Novelty confidence is moderate: the argument appears unrecorded, but it is an immediate overlooked consequence of the source’s own adjacent results.

Human review should verify that the norm in footnote 33 is the ambient operator norm and that Corollary 2.3.12 is onto the full Q_H appearing in the question. Both are explicit in the reproduced source pages.

References

- [1] F. vom Ende, *Reachability in Controlled Markovian Quantum Systems: An Operator-Theoretic Approach*, arXiv:2012.03496v2, 2020.